

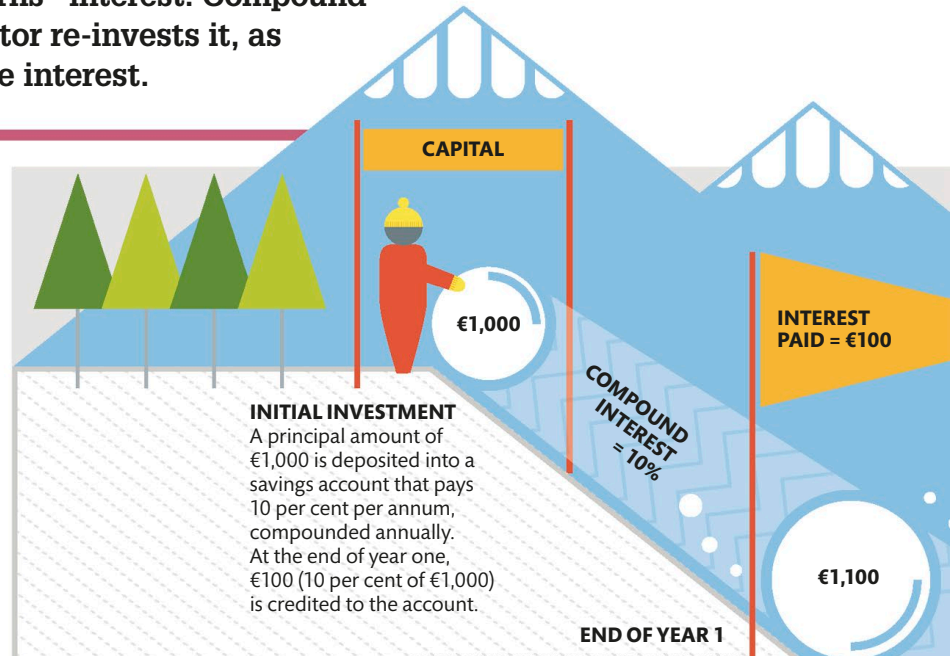


Interest and compound interest

When money is saved it “earns” interest. Compound interest accrues if the investor re-invests it, as opposed to withdrawing the interest.

The snowball effect

If a snowball is rolled down a hill, it gets bigger and bigger as it gathers more snow. The rate at which the snowball grows also increases as it rolls down the hill because there is a greater surface area for the snow to stick to. So, given enough time, a tiny snowball can become a giant one. Compound interest has been described as a “snowball effect” as it works in much the same way, meaning that a small investment can provide bigger returns than an investment where the interest sum is paid out to the investor annually.



NEED TO KNOW

- **Principal amount** The original capital sum invested or borrowed.
- **Compounding frequency** The number of times that interest is added to the principal amount in one year. For example, if interest is added monthly, the compounding frequency is 12.
- **Effective interest rate (EIR)** Also referred to as annual equivalent rate (AER). Takes into account the number of compounding periods within a specific period of time, so can be used to compare financial products with different compounding frequency.



Compound interest formula

“A” is the final amount in the savings account after “T” years’ interest compounded “N” times, at interest rate “R” on the starting amount of “P”.

$$A = P \left(1 + \frac{R}{N} \right)^{NT}$$

Final amount

Rate of interest

Time in years

Principal (original) sum

Number of times per year interest is compounded

INVESTMENT GROWS

The savings account now has €1,100, then earns €110 (10 per cent of €1,100) interest in the second year. By the end of year two the account has a balance of €1,210.